

The Tree-of-Thought Approach to Prompt Engineering



Estimated time needed: 40 minutes

Learning objectives:

After completing this lab, you will be able to:

- Describe the tree of thoughts prompting approach and explain how it differs from linear or chain-of-thought prompting
- Apply the tree of thoughts approach to analyze a multi-step problem by generating and evaluating multiple solution paths
- Design original ToT-style prompts tailored to diverse problem contexts, ensuring clear reasoning branches and evaluation criteria

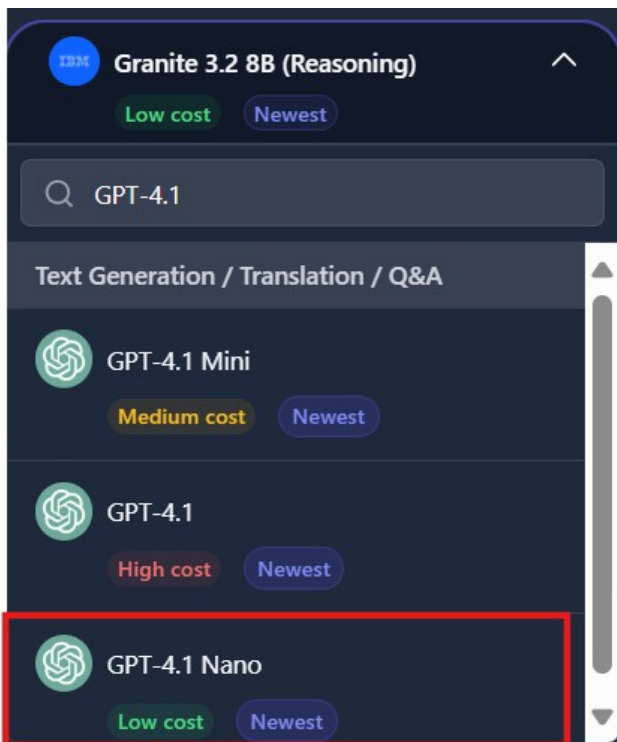
Introduction

The tree of thoughts (ToT) approach is a prompting strategy designed to enhance the reasoning abilities of large language models by structuring thinking as a tree-like process. Instead of generating a single linear response, the model is encouraged to explore multiple reasoning paths, evaluate them, and select the most promising one. This allows for more deliberate and interpretable decision-making, especially in complex or open-ended tasks.

By breaking a problem into intermediate steps and branching out into alternative ideas or strategies at each stage, ToT enables deeper exploration. It supports critical thinking, planning, and reflection by allowing paths to be compared and refined. The approach mirrors how humans often reason, by weighing different possibilities before converging on a conclusion—making it highly effective for creative, analytical, or ambiguous problems.

Choose the model:

To select the model, use the dropdown option from the top-left corner of the right pane for the text generation exercise and choose "GPT-4.1 Nano" model.



Examples of tree of thought approach (part 1)

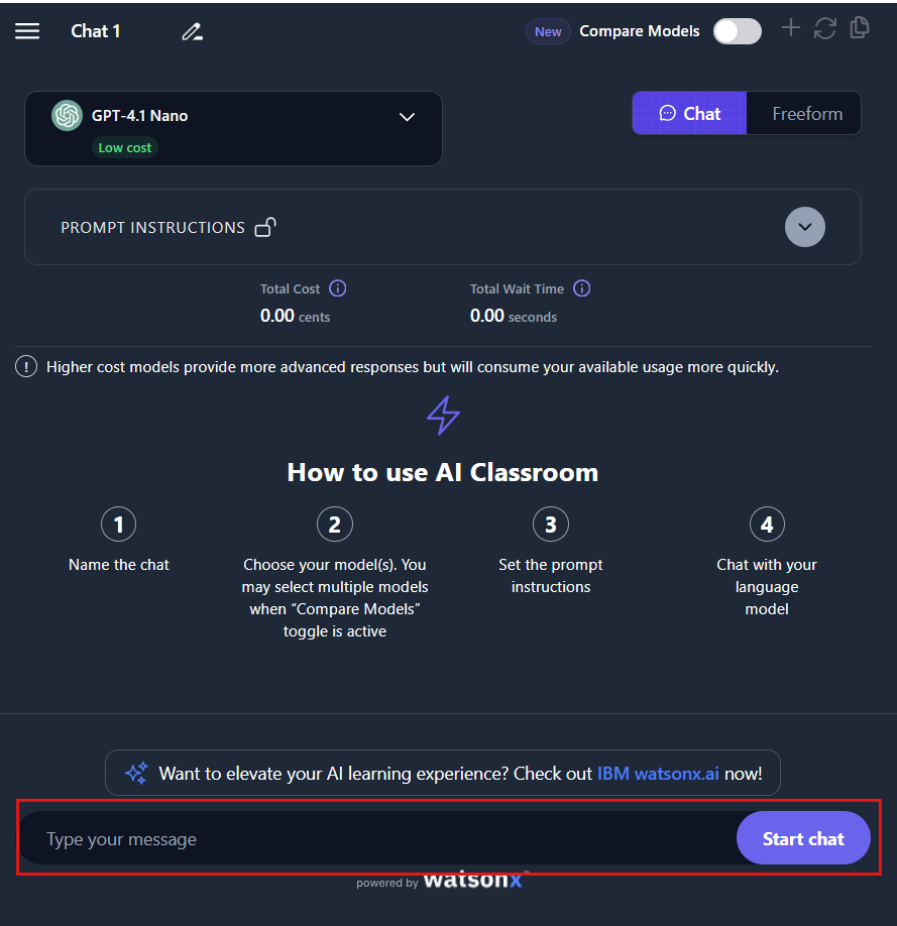
Let's look at an example which demonstrate the power of the tree of thought approach.

Example 1: Planning a fundraising event

Consider a scenario where you have been made in charge of a fundraiser being organized by your school and you are using a GenAI system to plan it out. You can query your GenAI system using the prompt below.

You are planning a school fundraising event.
Follow this structure:
List three different types of events (label them A, B, C).
For each event, list:
 a. Key benefits
 b. Likely challenges
 c. What resources would be needed
Compare the three events and choose the most feasible one. Explain why it is better than the others.

Copy this prompt and paste it in the "Type your message" block and press "Start chat" as shown in the image.



Here, the tree of thoughts (ToT) approach is used to systematically explore multiple event ideas—each developed into a thought branch with sub-criteria like benefits, challenges, and required resources. This example shows how ToT encourages **structured divergence** before making a decision. Rather than settling on the first plausible idea, the model is prompted to **generate and evaluate multiple possibilities**.

This structured branching not only leads to a more informed and creative outcome (e.g., selecting a student-led talent show over conventional options) but also provides **transparency in reasoning**. It demonstrates the strength of ToT in helping models (and humans) avoid tunnel vision, compare trade-offs, and make decisions grounded in thoughtful analysis. For learners, it highlights how ToT supports **deliberate thinking over default thinking**, which is crucial in open-ended or planning-heavy tasks.

The response may look as follows:

Fundraising Event Options

Event A: Bake Sale

Key Benefits:

- Low setup costs
- Appeals to a wide audience
- Quick to organize

Likely Challenges:

- Limited profit margins
- Requires volunteers for baking and sales
- Perishable items may lead to waste

Resources Needed:

- Baking supplies
- Tables and signage
- Volunteers for baking and selling

Event B: Car Wash**Key Benefits:**

- Simple setup
- Can be held outdoors
- Good for community involvement

Likely Challenges:

- Weather-dependent
- Water and cleaning supplies required
- Need for volunteers and possibly water disposal plans

Resources Needed:

- Buckets, sponges, soap, hoses
- Signage and location permissions
- Volunteers

Event C: Talent Show**Key Benefits:**

- Showcases students' talents
- Can attract a larger audience
- Potential for ticket sales and concessions

Likely Challenges:

- Requires coordination and rehearsals
- Need for a venue and technical equipment
- Possible costs for equipment and promotion

Resources Needed:

- Venue (auditorium or gym)
- Sound and lighting equipment
- Volunteers for organization and ticketing

Most Feasible Event

Considering setup complexity, resource requirements, and potential for quick fundraising, **Event A: Bake Sale** appears most feasible. It involves minimal costs, can be organized rapidly, and leverages existing community support. While the profit margins are lower, its simplicity and lower risk make it ideal for a school setting with limited planning time and resources.

Examples of tree of thought approach (part 2)

We can apply the tree of thought approach for exploring more personal requirements as well. Consider the following example.

Example 2: Designing a family meal plan on a budget

You have decided to take a renewed approach toward improving the health of your family, but are feeling confused between different approaches of eating healthy. You can use the model to design a weekly meal plan for your family based on one of three different approaches that appeal to you. Consider the following prompt.

You are helping a family of four plan healthy meals for the week on a tight budget. Follow this structure:
Propose three different meal plan strategies (label them A, B, C). For example:

- A: Vegetarian-heavy meals
- B: Mix of bulk grains and frozen foods
- C: One-pot meals to reduce prep

For each strategy, list:

- a. Estimated cost per day
- b. Nutritional strengths
- c. Convenience level (prep time, cleanup)

Compare the three strategies and choose the best one. Explain why it's the most practical based on cost, nutrition, and convenience.

This example highlights how the tree of thoughts (ToT) approach supports practical decision-making when balancing multiple priorities—in this case, cost, nutrition, and convenience. Instead of generating a single solution, the model is guided to explore three distinct planning strategies, such as vegetarian meals, bulk ingredient use, and one-pot cooking. Each strategy is then broken down into specific factors: daily cost, nutritional value, and ease of preparation.

This example shows the power of ToT in everyday planning by encouraging structured exploration, not just quick answers. By comparing each option against real-world constraints, the model can reason more deliberately and arrive at a well-justified recommendation. For learners, it reinforces how ToT helps in thinking beyond the obvious, evaluating trade-offs, and making choices that fit real needs—even in something as routine as planning meals.

The expected output will be as follows.

Here are three distinct meal plan strategies tailored for a family of four on a tight budget, each with its own advantages:

Strategy A: Vegetarian-Heavy Meals

Estimated Cost per Day: \$15

Nutritional Strengths: Rich in fiber, vitamins, and plant-based proteins; supports heart health.

Convenience Level: Moderate; involves preparing multiple components like grains, vegetables, and legumes, but recipes are straightforward.

Strategy B: Mix of Bulk Grains and Frozen Foods

Estimated Cost per Day: \$12

Nutritional Strengths: Cost-effective with a good balance of carbs, some proteins, and vitamins from frozen vegetables.

Convenience Level: High; bulk grains (like rice, oats) and frozen vegetables reduce prep time and waste.

Strategy C: One-Pot Meals

Estimated Cost per Day: \$14

Nutritional Strengths: Efficient use of ingredients, often includes vegetables, grains, and proteins in one dish, ensuring balanced nutrition.

Convenience Level: Very high; minimal prep and cleanup, ideal for busy days.

Comparison and Best Choice

Among these, Strategy B (Mix of Bulk Grains and Frozen Foods) strikes the best balance. It offers the lowest daily cost, maintains good nutritional value with minimal prep time, and reduces waste. While Strategy A is healthier in some aspects, it may require more time and effort. Strategy C is highly convenient but slightly more expensive and potentially less varied.

Conclusion: For a family prioritizing budget, nutrition, and ease, Strategy B is the most practical choice.

Try yourself

Try drafting a prompt for the following scenarios which makes the model use a tree of thought approach towards generating a response.

1. Crafting a plot twist for a short story

Domain: Creative writing

Scenario:

You're writing a mystery short story and want to include a compelling twist at the end that surprises the reader but still makes logical sense.

Learner task:

Design a prompt that encourages the model to:

- Explore three different twist ideas (e.g., unreliable narrator, secret identity, and fake crime)
- For each, evaluate originality, how well it fits the story so far, and how easy it is to foreshadow
- Choose the most satisfying twist and explain why it enhances the story's impact

► [Click here for a sample prompt](#)

2. Fixing a decline in customer satisfaction

Domain: Business/customer support

Scenario:

A company notices that its customer satisfaction score has dropped significantly in the past quarter. They want to understand why and how to fix it.

Learner task:

Write a ToT-style prompt that guides the model to:

- Suggest three likely causes (e.g., delayed support, product bugs, and unclear policies)
- Outline supporting data to look for, the impact on users, and a potential fix for each cause
- Recommend the root cause and the highest-priority fix

► [Click here for a sample prompt](#)

3. Planning a gap year abroad

Domain: Personal planning/travel

Scenario:

A student is taking a gap year and wants to make the most of it through travel, volunteering, or learning new skills, but can only choose one main focus.

Learner task:

Design a prompt that asks the model to:

- Outline three different approaches (e.g., travel across Asia, join a conservation program, take a coding bootcamp)
- For each, list potential costs, personal growth opportunities, and long-term benefits
- Select the most meaningful plan based on the student's goals

► [Click here for a sample prompt](#)

Conclusion

Congratulations on completing this lab, where you learned how to use the tree of thoughts prompting approach to break down complex problems into structured reasoning paths. You explored real-world scenarios, compared multiple solutions, and practiced designing thoughtful prompts to guide more deliberate decision-making.

Author(s)

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